

Scenario A: Correct Client Certificate

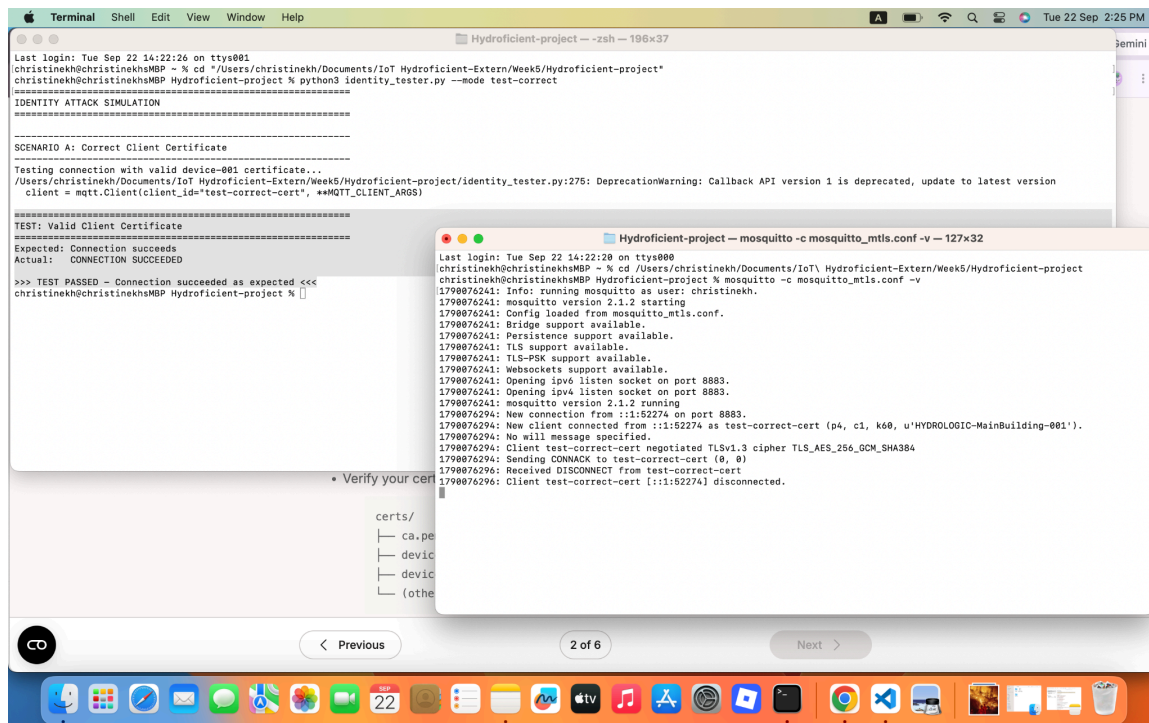
Goal: Verify that normal mTLS connections work.

Field	Your Result
Connection outcome	☒ Succeeded
Error message (if any)	None
Expected outcome	Connection succeeds
Test result	☒ PASS []

Analysis: Did the connection succeed as expected?

Yes. The connection succeeded as expected because the device presented a valid client certificate signed by the trusted CA.

Screenshot:



```
Terminal Shell Edit View Window Help
Hydroficient-project -- zsh -- 196x37

Last login: Tue Sep 22 14:22:26 on ttys001
christinekh@christinekhMBP ~ % cd "/Users/christinekh/Documents/IoT Hydroficient-Extern/Week5/Hydroficient-project"
christinekh@christinekhMBP Hydroficient-project % python3 identity_tester.py --mode test-correct

IDENTITY ATTACK SIMULATION
=====

SCENARIO A: Correct Client Certificate
-----
Testing connection with valid device-001 certificate...
/Users/christinekh/Documents/IoT Hydroficient-Extern/Week5/Hydroficient-project/identity_tester.py:275: DeprecationWarning: Callback API version 1 is deprecated, update to latest version
  client = mqtt.Client(client_id="test-correct-cert", **MQTT_CLIENT_ARGS)

=====
TEST: Valid Client Certificate
=====
Expected: Connection succeeds
Actual: CONNECTION SUCCEEDED

>>> TEST PASSED - Connection succeeded as expected <<<
christinekh@christinekhMBP Hydroficient-project %
```

```
Hydroficient-project -- mosquito -c mosquito_mtls.conf -v -- 127x32

Last login: Tue Sep 22 14:22:20 on ttys000
christinekh@christinekhMBP ~ % cd "/Users/christinekh/Documents/IoT Hydroficient-Extern/Week5/Hydroficient-project"
christinekh@christinekhMBP Hydroficient-project % mosquito -c mosquito_mtls.conf -v
1790076241: Info: running mosquito as user: christinekh.
1790076241: mosquito version 2.1.2 starting
1790076241: Config loaded from mosquito_mtls.conf.
1790076241: Bridge support available.
1790076241: Persistence support available.
1790076241: TLS support available.
1790076241: TLS-PSK support available.
1790076241: Websockets support available.
1790076241: Opening ipv6 listen socket on port 8883.
1790076241: Opening ipv4 listen socket on port 8883.
1790076241: mosquito version 2.1.2 running
1790076294: New connection from ::1:52274 on port 8883.
1790076294: New client connected from ::1:52274 as test-correct-cert (p4, c1, k60, u'HYDROLOGIC-MainBuilding-001').
1790076294: No will message specified.
1790076294: Client test-correct-cert negotiated TLSv1.3 cipher TLS_AES_256_GCM_SHA384
1790076294: Sending CONNACK to test-correct-cert (0, 0)
1790076296: Received DISCONNECT from test-correct-cert
1790076296: Client test-correct-cert (::1:52274) disconnected.
```

Scenario B: No Client Certificate

Goal: Verify that the broker rejects devices without certificates.

Field	Your Result
Connection outcome	☒ Failed
Error message (if any)	Connection failed with rc=-1
Expected outcome	Connection rejected
Test result	☒ Pass

Analysis: What specific error did you see? This error proves that a rogue device cannot connect just by having network access.

The connection was rejected because the device did not provide a client certificate. The broker requires client certificates for mTLS, so having network access alone is not enough to connect. The broker log can provide the more specific TLS error: **peer did not return a certificate**.

Screenshot:

The screenshot shows two terminal windows. The left window displays the output of a Python script named 'identity_tester.py' which tests a connection without a client certificate. It shows a deprecation warning and a 'Connection FAILED' error with rc=-1. The right window shows the Mosquitto broker logs, which include information about the broker's startup, supported features, and the specific error received from the client: 'error:0A0000C7:SSL routines:peer did not return a certificate'.

```
Hydroficient-project -- zsh -- 102x48
Last login: Tue Sep 22 14:22:28 on ttys002
christinekh@christinekhMBP ~ % cd /Users/christinekh/Documents/IoT\ Hydroficient-Extern/Week5/Hydroficient-project
christinekh@christinekhMBP Hydroficient-project % python3 identity_tester.py --mode test-no-cert
=====
IDENTITY ATTACK SIMULATION
=====
SCENARIO B: No Client Certificate
=====
Testing connection WITHOUT any client certificate...
/Users/christinekh/Documents/IoT\ Hydroficient-Extern/Week5/Hydroficient-project/identity_tester.py:314
: DeprecationWarning: Callback API version 1 is deprecated, update to latest version
client = mqtt.Client(client_id="test-no-cert", **MQTT_CLIENT_ARGS)
=====
TEST: No Client Certificate
=====
Expected: Connection rejected
Actual: CONNECTION FAILED
Error: Connection failed with rc=-1

>>> TEST PASSED - Connection was correctly rejected <<<
christinekh@christinekhMBP Hydroficient-project %
```

```
Hydroficient-project -- mosquitto -c mosquitto_mtls.conf -v -- 98x36
Last login: Tue Sep 22 14:26:12 on ttys000
christinekh@christinekhMBP ~ % cd /Users/christinekh/Documents/IoT\ Hydroficient-Extern/Week5/Hydroficient-project
christinekh@christinekhMBP Hydroficient-project % mosquitto -c mosquitto_mtls.conf -v
1790076453: Info: running mosquitto as user: christinekh.
1790076453: mosquitto version 2.1.2 starting
1790076453: Config loaded from mosquitto_mtls.conf.
1790076453: Bridge support available.
1790076453: Persistence support available.
1790076453: TLS support available.
1790076453: TLS-PSK support available.
1790076453: Websockets support available.
1790076453: Opening ipv6 listen socket on port 8883.
1790076453: Opening ipv4 listen socket on port 8883.
1790076453: mosquitto version 2.1.2 running
1790076478: New connection from ::1:52282 on port 8883.
1790076478: OpenSSL Error while trying to get the error[0]: error:0A0000C7:SSL routines:peer did not return a certificate
1790076478: Client ::1 [::1:52282] disconnected: Protocol error.
1790076479: New connection from ::1:52285 on port 8883.
1790076479: OpenSSL Error while trying to get the error[0]: error:0A0000C7:SSL routines:peer did not return a certificate
1790076479: Client ::1 [::1:52285] disconnected: Protocol error.
```

Scenario C: Certificate from Wrong CA

Goal: Verify that only certificates signed by your CA are accepted.

Field	Your Result
Connection outcome	☒ Failed

Error message (if any)	Connection failed with rc=-1
Expected outcome	Connection rejected
Test result	☑ PASS

Analysis: How does this differ from Scenario B? Both are rejected, but for different reasons.

This differs from Scenario B because the device **did provide a certificate**, but the certificate was signed by a different CA that the broker does not trust. In Scenario B, no client certificate was provided at all. In both cases, the broker rejects the connection.

Screenshot:

```

Last login: Tue Sep 22 14:26:16 on ttys001
christinekh@christinekhMBP ~ % cd /Users/christinekh/Documents/IoT\ Hydroficient-Extern\Week5\Hydroficient-project
christinekh@christinekhMBP ~ % python3 identity_tester.py --mode test-wrong-ca
=====
IDENTITY ATTACK SIMULATION
=====
SCENARIO C: Certificate from Wrong CA
-----
Generating rogue CA and client certificate...
Done. Rogue CA and device certificate created.
Testing connection with certificate signed by DIFFERENT CA...
/Users/christinekh/Documents/IoT Hydroficient-Extern/Week5/Hydroficient-project/identity_tester.py:359: DeprecationWarning: Callback API version 1 is deprecated, update to latest version
  client = mqtt.Client(client_id="test-wrong-ca", **MQTT_CLIENT_ARGS)
=====
TEST: Wrong CA Certificate
-----
Expected: Connection rejected
Actual: CONNECTION FAILED
Error: Connection failed with rc=-1

>>> TEST PASSED - Connection was correctly rejected <<<
christinekh@christinekhMBP Hydroficient-project %

```

```

Last login: Tue Sep 22 14:26:12 on ttys001
christinekh@christinekhMBP ~ % cd /Users/christinekh/Documents/IoT\ Hydroficient-Extern\Week5\Hydroficient-project
christinekh@christinekhMBP ~ % mosquitto -c mosquitto_mtls.conf -v
1790076453: Info: running mosquitto as user: christinekh.
1790076453: mosquitto version 2.1.2 starting
1790076453: Config loaded from mosquitto_mtls.conf.
1790076453: Bridge support available.
1790076453: Persistence support available.
1790076453: TLS support available.
1790076453: TLS-PSK support available.
1790076453: Websockets support available.
1790076453: Opening ipv6 listen socket on port 8883.
1790076453: Opening ipv4 listen socket on port 8883.
1790076453: mosquitto version 2.1.2 running
1790076478: New connection from ::1:52282 on port 8883.
1790076478: OpenSSL Error while trying to get the error[0]: error:0A00000C:SSL routines:peer did not return a certificate
1790076478: Client ::1 [::1:52282] disconnected: Protocol error.
1790076479: New connection from ::1:52285 on port 8883.
1790076479: OpenSSL Error while trying to get the error[0]: error:0A00000C:SSL routines:peer did not return a certificate
1790076479: Client ::1 [::1:52285] disconnected: Protocol error.
1790076753: New connection from ::1:52289 on port 8883.
1790076753: OpenSSL Error while trying to get the error[0]: error:0A000006:SSL routines:certificate verify failed
1790076753: Client ::1 [::1:52289] disconnected: Protocol error.
1790076754: New connection from ::1:52292 on port 8883.
1790076754: OpenSSL Error while trying to get the error[0]: error:0A000006:SSL routines:certificate verify failed
1790076754: Client ::1 [::1:52292] disconnected: Protocol error.

```

Scenario D: Expired Certificate

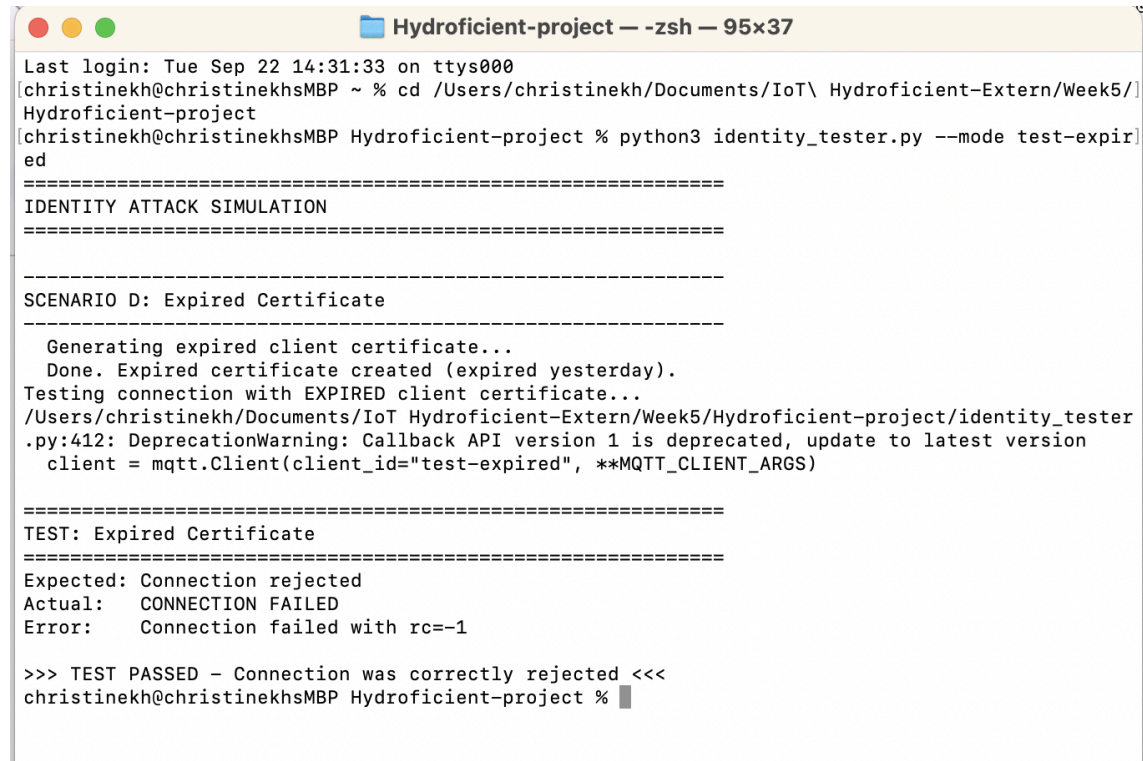
Goal: Verify that expired certificates are rejected.

Field	Your Result
Connection outcome	☑ Failed
Error message (if any)	Connection failed with rc=-1
Expected outcome	Connection rejected
Test result	☑ PASS

Analysis: Why is certificate expiration important for IoT security? What happens if a device certificate never expires?

Certificate expiration is important because it limits how long a device certificate can be trusted. If a certificate never expires, a compromised or stolen certificate could potentially remain valid indefinitely. Expiration forces certificates to be renewed and limits the lifetime of a compromised credential.

Screenshot:



```
Last login: Tue Sep 22 14:31:33 on ttys000
christinekh@christinekhsMBP ~ % cd /Users/christinekh/Documents/IoT\ Hydroficient-Extern/Week5/
Hydroficient-project
christinekh@christinekhsMBP Hydroficient-project % python3 identity_tester.py --mode test-expired
=====
IDENTITY ATTACK SIMULATION
=====

-----
SCENARIO D: Expired Certificate
-----

Generating expired client certificate...
Done. Expired certificate created (expired yesterday).
Testing connection with EXPIRED client certificate...
/Users/christinekh/Documents/IoT Hydroficient-Extern/Week5/Hydroficient-project/identity_tester
.py:412: DeprecationWarning: Callback API version 1 is deprecated, update to latest version
  client = mqtt.Client(client_id="test-expired", **MQTT_CLIENT_ARGS)

=====
TEST: Expired Certificate
=====
Expected: Connection rejected
Actual: CONNECTION FAILED
Error: Connection failed with rc=-1

>>> TEST PASSED - Connection was correctly rejected <<<
christinekh@christinekhsMBP Hydroficient-project %
```

Summary Worksheet

Complete this summary of your identity attack simulations:

Scenario	What You Tried	Expected	Actual	Pass/Fail
A: Correct cert	Connect with valid device-001 cert	Success	Connection succeeded	PASS

B: No cert	Connect without client certificate	Rejected	Connection failed	PASS
C: Wrong CA	Connect with cert from different CA	Rejected	Connection failed	PASS
D: Expired	Connect with expired certificate	Rejected	Connection failed	PASS

The Full Test Screenshot:

```
Last login: Tue Sep 22 14:35:29 on ttys000
christinekh@christinekhMBP ~ % cd /Users/christinekh/Documents/IoT\ Hydroficient-Extern/Week5/Hydroficient-project
christinekh@christinekhMBP Hydroficient-project % python3 identity_tester.py --mode all
=====
IDENTITY ATTACK SIMULATION SUITE
The Grand Marina Hotel - mTLS Testing
=====

-----
SCENARIO A: Correct Client Certificate
-----
Testing connection with valid device-001 certificate...
/Users/christinekh/Documents/IoT Hydroficient-Extern/Week5/Hydroficient-project/identity_tester.py:275: DeprecationWarning: Callback API version 1 is deprecated, update to latest version
  client = mqtt.Client(client_id="test-correct-cert", **MQTT_CLIENT_ARGS)

=====
TEST: Valid Client Certificate
=====
Expected: Connection succeeds
Actual:  CONNECTION SUCCEEDED

>>> TEST PASSED - Connection succeeded as expected <<<

-----
SCENARIO B: No Client Certificate
-----
Testing connection WITHOUT any client certificate...
/Users/christinekh/Documents/IoT Hydroficient-Extern/Week5/Hydroficient-project/identity_tester.py:314: DeprecationWarning: Callback API version 1 is deprecated, update to latest version
  client = mqtt.Client(client_id="test-no-cert", **MQTT_CLIENT_ARGS)

=====
TEST: No Client Certificate
=====
Expected: Connection rejected
Actual:  CONNECTION FAILED
Error:   Connection failed with rc=-1

>>> TEST PASSED - Connection was correctly rejected <<<

-----
SCENARIO C: Certificate From Wrong CA
-----
Testing connection with certificate signed by DIFFERENT CA...
/Users/christinekh/Documents/IoT Hydroficient-Extern/Week5/Hydroficient-project/identity_tester.py:359: DeprecationWarning: Callback API version 1 is deprecated, update to latest version
  client = mqtt.Client(client_id="test-wrong-ca", **MQTT_CLIENT_ARGS)

=====
TEST: Wrong CA Certificate
=====
Expected: Connection rejected
Actual:  CONNECTION FAILED
Error:   Connection failed with rc=-1

>>> TEST PASSED - Connection was correctly rejected <<<

-----
SCENARIO D: Expired Certificate
-----
Testing connection with EXPIRED client certificate...
/Users/christinekh/Documents/IoT Hydroficient-Extern/Week5/Hydroficient-project/identity_tester.py:412: DeprecationWarning: Callback API version 1 is deprecated, update to latest version
  client = mqtt.Client(client_id="test-expired", **MQTT_CLIENT_ARGS)

=====

=====
TEST: Expired Certificate
=====
Expected: Connection rejected
Actual:  CONNECTION FAILED
Error:   Connection failed with rc=-1

>>> TEST PASSED - Connection was correctly rejected <<<

=====
TEST SUMMARY
=====
A: Correct cert: PASS
B: No cert: PASS
C: Wrong CA: PASS
D: Expired: PASS

Total: 4/4 tests passed
christinekh@christinekhMBP Hydroficient-project %
```